ERDŐS SIMILARITY FOR DISTANCE SETS: FINITE PATTERNS, A PINNED OBSTRUCTION, AND AN OPEN WINDOW

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ABSTRACT. For a compact set $E \subset \mathbb{R}^d$ let $\Delta(E) = \{|x-y| : x, y \in E\}$ be its distance set. We ask whether, for $d \geq 2$ and $\dim_{\mathrm{H}} E > d/2$, the distance set must contain a nontrivial affine copy $x + s\{2^{-n} : n \geq 1\}$ of the dyadic sequence, a distance-set version of the Erdős similarity problem. We combine two recent results announced by OpenAI: the Falconer distance theorem in all dimensions and the dyadic case of the Erdős similarity conjecture. Assuming these, we show: (A) $\Delta(E)$ contains affine copies of every finite set, for a positive-measure family of translations at every small scale. (B) For every $d \geq 1$ there is a compact set $K \ni 0$ of nearly full volume in the unit ball of $\mathbb{R}^d$ whose distance set from the origin contains no affine copy of the dyadic sequence, while $\Delta(K)$ does contain one and, for $d \geq 2$, so does the distance set from every other point. (C) A counterexample must have $\Delta(E)$ of positive measure but empty interior, so by the Mattila–Sjölin theorem the question is open exactly for $d/2 < \dim_{\mathrm{H}} E \leq (d+1)/2$. A positive answer would follow if $\Delta(E)$ had nonempty interior in this range, itself an open problem; a negative answer would, to our knowledge, give the first sets of dimension $> d/2$ whose distance sets have empty interior. Every lemma, proposition and theorem of this note has a Lean 4 proof with Mathlib. The OpenAI results and the Mattila–Sjölin theorem enter as explicit hypotheses, and we checked with the Lean kernel that OpenAI's Lean proofs and a separate formalization of the Mattila–Sjölin theorem prove exactly these hypotheses. So Theorems A and B, Proposition C and the window are machine-checked down to the standard axioms. The spherical maximal theorem, which is not formalized, is used only in one clearly marked remark.

1. INTRODUCTION

1.1. **Two problems about large sets.** A set $P \subset \mathbb{R}$ is *measure universal* if every measurable $S \subset \mathbb{R}$ of positive Lebesgue measure contains a nontrivial affine copy $x + sP$ ($x \in \mathbb{R}$, $s \neq 0$). Steinhaus' theorem [21] implies that every finite set is measure universal. Erdős conjectured [5] that no infinite set is. The conjecture is known for slowly decaying sequences [7] and for many structured sets [3, 11]. The geometric sequences $\{q^n\}$ remained open, above all the dyadic sequence

$$\mathcal{D} = \{2^{-n} : n \geq 1\};$$

deciding whether $\mathcal{D}$ is measure universal has been called "the major bottleneck of the problem" [10]. OpenAI has released a collection of manuscripts, dated September and October 2026, produced by an internal AI model. It includes a proof that $\mathcal{D}$ is not measure universal [14] and an extension to every ratio $q \in (0,1)$ [16].

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Falconer's distance problem [8] asks for the smallest α such that every compact $E \subset \mathbb{R}^d$ with $\dim_{\mathrm{H}} E > \alpha$ has a distance set

$$\Delta(E) = \{|x - y| : x, y \in E\}$$

of positive Lebesgue measure. Falconer showed that $d/2$ would be optimal and conjectured it. Partial progress includes [23, 6, 9]. The same OpenAI release contains a proof of the full conjecture in every dimension $d \geq 2$ [15]. Mattila and Sjölin [13] showed that $\dim_{\mathrm{H}} E > (d+1)/2$ even forces $\Delta(E)$ to contain an interval.

1.2. **The question.** The Falconer theorem says that $\Delta(E)$ is large in measure, while the dyadic theorem says that being large in measure is not enough to contain $\mathcal{D}$. Distance sets, however, have additional multi-scale structure: $\Delta(E) \supseteq \Delta(E \cap B)$ for every ball B. In particular (Lemma 3.3), if $\dim_{\mathrm{H}} E > d/2$ then $\Delta(E) \cap [0, r]$ has positive measure for every $r > 0$. This alone does not force a copy of $\mathcal{D}$ (Proposition 3.4), but it motivates:

Question 1.1 (Erdős similarity for distance sets)**.** Let $d \geq 2$ and let $E \subset \mathbb{R}^d$ be compact with $\dim_{\mathrm{H}} E > d/2$. Must $\Delta(E)$ contain a nontrivial affine copy of $\mathcal{D}$?

We could not find this question, or any study of Erdős-type patterns inside distance sets, in the literature; it is not discussed in the recent survey [10].

1.3. **Results.** Theorem A assumes the Falconer theorem (Theorem 2.1), and Theorem B assumes the dyadic avoidance theorem (Theorem 2.2); both inputs are stated precisely in Section 2.

Theorem A (Finite patterns)**.** *Assume Theorem 2.1. Let $d \geq 2$, let $E \subset \mathbb{R}^d$ be compact with $\dim_{\mathrm{H}} E > d/2$, and let $F \subset \mathbb{R}$ be finite. There is $\delta > 0$ such that for every $|s| < \delta$,*

$$\mathcal{L}^1(\{x \in \mathbb{R} : x + sF \subseteq \Delta(E)\}) > 0.$$

In particular $\Delta(E)$ contains affine copies $x + sF$ with $0 < s < \delta'$ and with $-\delta' < s < 0$, for every $\delta' > 0$.

Theorem B (A pinned obstruction)**.** *Assume Theorem 2.2. Let $d \geq 1$ and $\eta \in (0, 1)$. There is a compact set K in the closed unit ball of $\mathbb{R}^d$ with $0 \in K$ and*

$$\mathcal{L}^d(K) \;\geq\; (1 - \eta)^d \, \mathcal{L}^d(B(0, 1)) > 0, \qquad \text{hence } \dim_{\mathrm{H}} K = d,$$

such that the pinned distance set $\Delta_0(K) = \{|y| : y \in K\}$ contains no nontrivial affine copy of $\mathcal{D}$, whereas the full distance set $\Delta(K)$ contains one. If $d \geq 2$, then for every $p \in \mathbb{R}^d \setminus \{0\}$ the pinned distance set $\Delta_p(K) = \{|p - y| : y \in K\}$ contains a nondegenerate interval, hence an affine copy of $\mathcal{D}$.

So pinned distance sets are not universal hosts for $\mathcal{D}$, even from a pin inside a set of full dimension and nearly full volume in the ball. For $d \geq 2$, however, the obstruction sits at one single point, and in measure it cannot be larger: in a set of positive measure, almost every pin sees all small distances (Remark 4.7). Theorem B holds verbatim for every ratio $q \in (0, 1)$ in place of $1/2$, assuming the corresponding avoidance statement (Theorem 4.6).

Proposition C. *Let $d \geq 2$, let $E \subset \mathbb{R}^d$ be compact, and let $q \in (0, 1)$.*

(i) *If $\Delta(E)$ has nonempty interior, it contains an affine copy of every bounded set $P \subset \mathbb{R}$, in particular of $\{q^n : n \geq 1\}$.*
(ii) *This applies whenever $\mathcal{L}^d(E) > 0$.*
(iii) *Assume Theorem 2.1. If $\dim_{\mathrm{H}} E > d/2$ and $\Delta(E)$ contains no affine copy of $\{q^n : n \geq 1\}$, then $\mathcal{L}^1(\Delta(E)) > 0$, $\Delta(E)$ has empty interior, and $\mathcal{L}^d(E) = 0$.*

Combined with the Mattila–Sjölin theorem, a counterexample to Question 1.1 must have $d/2 < \dim_{\mathrm{H}} E \le (d+1)/2$ (Remark 5.1), so the question is open exactly in this window: we know of no result that gives nonempty interior of $\Delta(E)$ below the threshold $(d+1)/2$, and Borges, Foster, Ou and Palsson note that "neither of the results on nonempty interior has been improved since the original papers" [1], meaning those of Mattila–Sjölin and of Peres–Schlag [18]. Proposition C also relates Question 1.1 to a known open problem: does $\dim_{\mathrm{H}} E > d/2$ already force $\Delta(E)$ to have nonempty interior (see [12] for background)? Opinions differ. For the pinned version, Borges, Foster, Ou and Palsson write that "it is widely believed that the true threshold should be $d/2$" [1], while Palsson and Romero Acosta point to evidence from finite fields that the two thresholds may differ [17]. A positive answer to that problem answers Question 1.1 positively, by Proposition C(i). Conversely, by Proposition C(iii), a negative answer to Question 1.1 would give the first known compact sets with $\dim_{\mathrm{H}} E > d/2$ whose distance sets have empty interior: according to [17], the only known sharpness examples for the Mattila–Sjölin theorem are those for the Falconer problem, which have dimension at most $d/2$. Section 6 lists further questions.

1.4. **Formalization and status of the inputs.** *Every lemma, proposition and theorem of this note has a proof in Lean 4* [4] *with Mathlib* [22]. For the inputs, Theorems 2.1, 2.2 and 2.4, these are OpenAI's proofs and a separate formalization of the Mattila–Sjölin theorem [19], which we re-checked; everything else is proved here. The Lean name is given next to each statement and in Table 1. External results enter our proofs only as explicit hypotheses, never as axioms:

(1) the Falconer theorem [15] (Theorem 2.1), used in Theorem A, Lemma 3.3, Proposition 3.2 and Proposition C(iii);
(2) the dyadic avoidance theorem [14] (Theorem 2.2), used in Theorem B and Proposition 3.4;
(3) the geometric avoidance statement for a ratio $q \neq 1/2$ [16] (Statement 2.3), used only in Theorem 4.6;
(4) the Mattila–Sjölin theorem [13] (Theorem 2.4), used only in Remark 5.1.

Remark 4.7 and item (2) of Section 6 cite further results from harmonic analysis (the spherical maximal theorem and pinned versions of the Mattila–Sjölin theorem). They are not formalized, and nothing else depends on them.

Inputs (1)–(3) come from an AI-generated, unrefereed manuscript collection whose authors state that "some of the unformalized results could have issues". Inputs (1) and (2) are among the formalized ones: OpenAI provides Lean proofs of both. We rebuilt the relevant part of that library ourselves and checked with the Lean kernel that these proofs establish exactly our hypotheses (Section 7). Input (4) is a classical peer-reviewed theorem. Its Lean proof [19] was written separately, also with AI assistance, and reuses some of OpenAI's Lean modules; again we checked with the kernel that it proves exactly Theorem 2.4. Consequently Theorems A and B, Propositions 3.2, 3.4 and C, Remark 5.1 and all lemmas are machine-checked down to Lean's standard axioms. Only input (3) is not: it is one of the unformalized results and has no Lean proof. A formal proof certifies the statement as formalized; it does not replace independent mathematical review of [15, 14].

2. Notation and inputs

$\mathcal{L}^d$ is Lebesgue measure on $\mathbb{R}^d$, $|\cdot|$ the Euclidean norm, $B(0,1)$ the open unit ball, and $\dim_{\mathrm{H}}$ Hausdorff dimension. For $E \subset \mathbb{R}^d$ and $p \in \mathbb{R}^d$ we write $\Delta(E) = \{|x-y| : x, y \in E\}$ and $\Delta_p(E) = \{|p-y| : y \in E\}$. An *affine copy* of $P \subset \mathbb{R}$ is a set $x + sP$ with $x \in \mathbb{R}$ and $s \neq 0$. For $q \in (0,1)$ let $\mathcal{G}_q = \{q^n : n \ge 1\}$, so $\mathcal{D} = \mathcal{G}_{1/2}$.

Theorem 2.1 (Falconer distance theorem [15]). *Let $d \geq 2$ and let $E \subset \mathbb{R}^d$ be compact with $\dim_{\mathrm{H}} E > d/2$. Then $\mathcal{L}^1(\Delta(E)) > 0$.* *[Lean: `FalconerStatement`]*

Theorem 2.2 (Dyadic avoidance [14]). *For every $\eta \in (0,1)$ there is a compact set $A \subseteq [0,1]$ with $\mathcal{L}^1(A) > 1-\eta$ such that for every $x \in \mathbb{R}$ and every $s \neq 0$ there is $n \geq 1$ with $x + s2^{-n} \notin A$.* *[Lean: `DyadicAvoidanceStatement`]*

Statement 2.3 (Geometric avoidance for the ratio q). *Statement* $\mathrm{GA}(q)$*: Theorem 2.2 with 2^{-n} replaced by q^n. For $q = 1/2$ it is Theorem 2.2; for all $q \in (0,1)$ it is claimed in [16], without a Lean proof.* *[Lean: `GeometricAvoidanceStatement`]*

Theorem 2.4 (Mattila–Sjölin [13]). *If $d \geq 2$ and $E \subset \mathbb{R}^d$ is compact with $\dim_{\mathrm{H}} E > (d+1)/2$, then $\Delta(E)$ has nonempty interior. A Lean proof is given in [19].* *[Lean: `MattilaSjolinStatement`]*

3. Finite patterns

The following is the standard Steinhaus-type argument (compare the introduction of [14]). We include it because Theorem A needs the version with a positive-measure set of translations at every small scale.

Lemma 3.1. *Let $S \subset \mathbb{R}$ be measurable with $0 < \mathcal{L}^1(S) < \infty$ and let $F \subset \mathbb{R}$ be finite. There is $\delta > 0$ such that $\mathcal{L}^1(\{x : x + sF \subseteq S\}) > 0$ for all $|s| < \delta$.* *[Lean: `eventually_volume_pattern_pos`]*

Proof. For $a \in F$ let $T_{a,s} = S - sa$. Translation is continuous in measure: $\mathcal{L}^1(T_{a,s} \triangle S) \to 0$ as $s \to 0$. Choose $\delta > 0$ with $\sum_{a\in F} \mathcal{L}^1(T_{a,s} \triangle S) < \mathcal{L}^1(S)$ for $|s| < \delta$. Since

$$S \subseteq \{x : x + sF \subseteq S\} \;\cup\; \bigcup_{a \in F} (S \setminus T_{a,s}),$$

we get $\mathcal{L}^1(S) \leq \mathcal{L}^1(\{x : x + sF \subseteq S\}) + \sum_{a\in F} \mathcal{L}^1(T_{a,s} \triangle S)$, and the claim follows. □

Proof of Theorem A. [Lean: `finite_pattern_distSet, exists_finite_pattern_distSet`] $\Delta(E)$ is the image of the compact set $E \times E$ under the continuous map $(x,y) \mapsto |x-y|$, so it is compact. In particular it is measurable with finite measure, and $\mathcal{L}^1(\Delta(E)) > 0$ by Theorem 2.1. Lemma 3.1 with $S = \Delta(E)$ gives the first claim. For the second, apply it with $s = \pm \min(\delta, \delta')/2$; a set of positive measure is nonempty. □

Proposition 3.2 (Planar form). *Under the assumptions of Theorem A, the set $\{(s,x) \in \mathbb{R}^2 : x + sF \subseteq \Delta(E)\}$ has positive planar Lebesgue measure.* *[Lean: `volume_pattern_pairs_pos`]*

Proof. The set is closed, hence measurable. By Tonelli's theorem its measure is $\int \mathcal{L}^1(\{x : x + sF \subseteq \Delta(E)\})\, ds$, and the integrand is positive for $|s| < \delta$. □

Lemma 3.3 (Distances at every scale). *Assume Theorem 2.1. If $d \geq 2$ and $E \subset \mathbb{R}^d$ is compact with $\dim_{\mathrm{H}} E > d/2$, then $\mathcal{L}^1(\Delta(E) \cap [0,r]) > 0$ for every $r > 0$.* *[Lean: `volume_distSet_inter_Icc_pos`]*

Proof. Cover E by finitely many closed balls $\bar{B}(c_i, r/2)$. Hausdorff dimension of a finite union is the maximum of the dimensions, so some compact piece $E_i = E \cap \bar{B}(c_i, r/2)$ has $\dim_{\mathrm{H}} E_i > d/2$. By Theorem 2.1, $\mathcal{L}^1(\Delta(E_i)) > 0$, and $\Delta(E_i) \subseteq \Delta(E) \cap [0,r]$. □

The property in Lemma 3.3 alone does not force a copy of $\mathcal{D}$:

Proposition 3.4 (Positive measure at every scale is not enough). *Assume Theorem 2.2. There is a compact set $S \subseteq [0,1]$ with $0 \in S$ and $\mathcal{L}^1(S \cap [0,r]) > 0$ for every $r > 0$ that contains no nontrivial affine copy of $\mathcal{D}$.* *[Lean: `exists_dyadicFree_pos_every_scale`]*

Proof. Let A be given by Theorem 2.2 for $\eta = 1/4$. Then $A' = A \cap [1/2, 1]$ has $\mathcal{L}^1(A') > 3/4 - 1/2 > 0$. Let $B_k = 8^{-k}A' \subseteq [8^{-k}/2, 8^{-k}]$ and $S = \{0\} \cup \bigcup_{k\geq 0} B_k$. Then $S = \bigcap_{N\geq 0}([0, 8^{-N}] \cup B_0 \cup \cdots \cup B_{N-1})$, so S is closed, hence compact. If $8^{-k} < r$, then $B_k \subseteq S \cap [0, r]$ and $\mathcal{L}^1(B_k) = 8^{-k}\mathcal{L}^1(A') > 0$.

If $y \in B_j$ and $z \in B_k$ with $j < k$, then $y \geq 8^{-j}/2 \geq 4 \cdot 8^{-k} \geq 4z$. So two points of $S \setminus \{0\}$ whose ratio lies strictly between $1/4$ and 4 lie in the same block. Suppose $p_n = x + s2^{-n} \in S$ for all $n \geq 1$, with $s \neq 0$. If $x = 0$, then all $p_n \neq 0$ and $p_n = 2p_{n+1}$, so by induction all p_n lie in the block of p_1. If $x \neq 0$, choose N with $|s|2^{-N} < |x|/8$. For $n \geq N$, p_n lies within $|x|/8$ of x. If $x < 0$ this contradicts $S \subseteq [0, 1]$; if $x > 0$, the ratio of any two such points lies in $(7/9, 9/7)$, so all p_n, $n \geq N$, lie in one block. In both cases there are k and $N \geq 1$ with $p_n \in B_k$ for all $n \geq N$. Then $8^k x + (8^k s2^{-N})2^{-m} = 8^k p_{N+m} \in A$ for all $m \geq 1$, contradicting the choice of A. □

Remark 3.5. Theorem A uses nothing about distance sets beyond compactness and positive measure. Its point is the contrast with Theorem B and Question 1.1: the finite-to-infinite gap of the Erdős similarity problem reappears inside Falconer distance sets.

4. The pinned obstruction

Lemma 4.1 (Adding the limit point)**.** *Let $q \in (0, 1)$ and let $A \subset \mathbb{R}$ contain no nontrivial affine copy of $\mathcal{G}_q$. Then neither does $A \cup \{0\}$.* *[Lean: `not_containsGeomCopy_insert_zero`]*

Proof. Suppose $x + sq^n \in A \cup \{0\}$ for all $n \geq 1$, with $s \neq 0$. The points $x + sq^m$, $m \geq 0$, are pairwise distinct. If none of them is 0, then $x + s\mathcal{G}_q \subseteq A$, a contradiction. Otherwise $x + sq^m = 0$ for exactly one $m \geq 0$. The tail $\{x + sq^{m+n} : n \geq 1\} = x + (sq^m)\mathcal{G}_q$ is then an affine copy of $\mathcal{G}_q$ avoiding 0, hence contained in A, again a contradiction. □

Lemma 4.2 (Radial sets)**.** *Let $d \geq 1$ and let $A \subseteq [0, 1]$ be compact with $\mathcal{L}^1(A) = c$. Then $K_A = \{y \in \mathbb{R}^d : |y| \in A\}$ satisfies $\mathcal{L}^d(K_A) \geq c^d\, \mathcal{L}^d(B(0, 1))$.* *[Lean: `volume_norm_preimage_ge`]*

Proof. In polar coordinates,

$$\mathcal{L}^d(K_A) = d\, \mathcal{L}^d(B(0, 1)) \int_A r^{d-1}\, dr.$$

Since $r \mapsto r^{d-1}$ is nondecreasing,

$$\begin{aligned}\int_A r^{d-1}dr - \int_0^c r^{d-1}dr &= \int_{A\setminus[0,c]} r^{d-1}dr - \int_{[0,c]\setminus A} r^{d-1}dr \\ &\geq c^{d-1}(\mathcal{L}^1(A \setminus [0, c]) - \mathcal{L}^1([0, c] \setminus A)) = c^{d-1}(\mathcal{L}^1(A) - c) = 0,\end{aligned}$$

so $\int_A r^{d-1}dr \geq c^d/d$. □

Lemma 4.3. *If $K \subset \mathbb{R}^d$ has $\mathcal{L}^d(K) > 0$, then $\dim_{\mathrm{H}} K = d$.* *[Lean: `dimH_eq_of_volume_pos`]*

Proof. For the sup-norm on $\mathbb{R}^d$, d-dimensional Hausdorff measure is a positive constant multiple of $\mathcal{L}^d$ (the constant depends on the normalisation of Hausdorff measure), and the Euclidean and sup norms are bi-Lipschitz equivalent. Hence $\mathcal{H}^d(K) > 0$ and $\dim_{\mathrm{H}} K \geq d$. The reverse inequality is trivial. □

Lemma 4.4 (Steinhaus)**.** *If $K \subset \mathbb{R}^d$ ($d \geq 1$) is compact with $\mathcal{L}^d(K) > 0$, then $[0, \varepsilon) \subseteq \Delta(K)$ for some $\varepsilon > 0$.* *[Lean: `exists_Ico_subset_distSet`]*

Proof. By Steinhaus' theorem $K - K \supseteq B(0, \varepsilon)$ for some $\varepsilon > 0$. For $0 \leq r < \varepsilon$ write $re_1 = x - y$ with $x, y \in K$; then $r = |x - y| \in \Delta(K)$. □

Lemma 4.5 (Other pins)**.** *Let $d \geq 2$, $\rho > 0$, $p \in \mathbb{R}^d \setminus \{0\}$, and let $K \subseteq \mathbb{R}^d$ contain the sphere $\{|y| = \rho\}$. Then $\Delta_p(K) \supseteq [\,||p| - \rho|, |p| + \rho\,]$.* *[Lean: `Icc_subset_pinnedDistSet`]*

Proof. The sphere is connected because $d \geq 2$, and the continuous function $y \mapsto |p - y|$ on it takes the values $||p| - \rho|$ at $\rho p/|p|$ and $|p| + \rho$ at $-\rho p/|p|$. By the intermediate value theorem it takes every value in between. □

Theorem 4.6 (Theorem B for a general ratio)**.** *Let $q \in (0, 1)$ and assume* GA(q) *(Statement 2.3). Then Theorem B holds with $\mathcal{D}$ replaced by $\mathcal{G}_q$.* *[Lean: `pinned_geometric_obstruction`]*

Proof. Let A be given by GA(q) for η, and put $K = \{y : |y| \in A \cup \{0\}\}$. The set $A \cup \{0\} \subseteq [0, 1]$ is compact, so K is compact and lies in the closed unit ball, and $0 \in K$. Since $K \supseteq K_A$ and $\mathcal{L}^1(A) > 1 - \eta$, Lemma 4.2 gives $\mathcal{L}^d(K) \geq (1 - \eta)^d \mathcal{L}^d(B(0, 1)) > 0$, and Lemma 4.3 gives $\dim_{\mathrm{H}} K = d$. Further, $\Delta_0(K) = \{|y| : y \in K\} \subseteq A \cup \{0\}$, which contains no affine copy of $\mathcal{G}_q$ by Lemma 4.1.

Any set $P \subseteq \mathbb{R}$ containing a nondegenerate interval $[a, b]$ contains the affine copy $a + (b - a)\mathcal{G}_q$. By Lemma 4.4, $\Delta(K)$ contains an interval. For $d \geq 2$ and $p \neq 0$, pick $\rho \in A$ with $\rho > 0$ (possible since $\mathcal{L}^1(A) > 0$); the sphere $\{|y| = \rho\}$ lies in K, so by Lemma 4.5 $\Delta_p(K)$ contains an interval of length $2 \min(\rho, |p|) > 0$. □

Proof of Theorem B. [Lean: `pinned_dyadic_obstruction`]
Theorem 4.6 with $q = 1/2$, where GA(1/2) is Theorem 2.2. □

Remark 4.7 (Bad pins form a null set; not formalized). Let $d \geq 2$ and let $E \subset \mathbb{R}^d$ be compact with $\mathcal{L}^d(E) > 0$. By the spherical maximal theorem (Stein [20] for $d \geq 3$, Bourgain [2] for $d = 2$), the averages of $\mathbf{1}_E$ over the spheres $\{y : |p - y| = r\}$ tend to $\mathbf{1}_E(p)$ as $r \to 0$ for almost every $p \in \mathbb{R}^d$. So for almost every $p \in E$ there is $r_0 > 0$ such that every sphere of radius $r < r_0$ about p meets E. Then $[0, r_0) \subseteq \Delta_p(E)$, and $\Delta_p(E)$ contains affine copies of every bounded set. Hence the pins $p \in E$ for which $\Delta_p(E)$ contains no affine copy of $\mathcal{D}$ form a Lebesgue null set. Theorem B shows that this set can be nonempty; for the set K constructed there and $d \geq 2$, it is $\{0\}$.

5. Where the question is open

Proof of Proposition C. [Lean: `distSet_containsAffineCopy_of_interior, distSet_containsAffineCopy_of_volume_pos, counterexample_properties`]
(i) A set with nonempty interior contains an interval $[a, b]$ with $a < b$. A bounded set P lies in some $[-M, M]$ with $M > 0$, and then $\frac{a+b}{2} + \frac{b-a}{2M} P \subseteq [a, b]$; and $\mathcal{G}_q \subseteq [0, 1]$ is bounded. (ii) follows from Lemma 4.4. (iii) $\mathcal{L}^1(\Delta(E)) > 0$ is Theorem 2.1; nonempty interior or $\mathcal{L}^d(E) > 0$ would give a copy of $\mathcal{G}_q$ by (i) and (ii). □

Remark 5.1 (Uses the Mattila–Sjölin theorem). If $d \geq 2$, $E \subset \mathbb{R}^d$ is compact and $\Delta(E)$ contains no affine copy of $\mathcal{G}_q$, then $\dim_{\mathrm{H}} E \leq (d + 1)/2$: otherwise $\Delta(E)$ has nonempty interior by Theorem 2.4, and Proposition C(i) applies. Hence a counterexample to Question 1.1, which has $\dim_{\mathrm{H}} E > d/2$ by assumption, lies in the window $d/2 < \dim_{\mathrm{H}} E \leq (d + 1)/2$, and by Proposition C(iii) its distance set has positive measure and empty interior. [Lean: `window_of_mattilaSjolin`]

6. Open questions

(1) Question 1.1 in the window $d/2 < \dim_{\mathrm{H}} E \leq (d + 1)/2$. A positive answer would follow from nonempty interior of $\Delta(E)$, but needs less; by Proposition 3.4 it needs multi-scale structure of distance sets beyond positive measure at every scale. A negative answer would

need a construction controlling all distances of a set of dimension $> d/2$ simultaneously. Such a set cannot contain a copy of itself scaled by $1/2$: if $f(E) \subseteq E$ for a similarity f of $\mathbb{R}^d$ with ratio $1/2$, then $\frac{1}{2}\Delta(E) \subseteq \Delta(E)$, so $t\,\mathcal{D} \subseteq \Delta(E)$ for every $t \in \Delta(E)$, and $t > 0$ exists as soon as E has two points. (This elementary observation is not formalized.)

(2) *All pins.* Is there a compact $E \subset \mathbb{R}^d$, $d \geq 2$, with $\dim_{\mathrm{H}} E > d/2$ such that $\Delta_p(E)$ contains no affine copy of $\mathcal{D}$ for *every* $p \in E$? Since $\Delta_p(E) \subseteq \Delta(E)$ for $p \in E$, every counterexample to Question 1.1 is such a set. Sets of positive measure are not, by Remark 4.7. Neither are sets of large dimension: Peres and Schlag [18] find a pin $p \in E$ for which $\Delta_p(E)$ has nonempty interior when $d \geq 3$ and $\dim_{\mathrm{H}} E > (d+2)/2$, and Borges, Foster, Ou and Palsson [1] do the same for $\dim_{\mathrm{H}} E > 7/4$ in the plane and $\dim_{\mathrm{H}} E > 12/5$ in $\mathbb{R}^3$. By Lemma 6.1 below, the question asks for a set whose complement contains, around every point of E, a sphere of radius r for some r in every affine copy of $\mathcal{D}$.

(3) The same questions for general infinite sets P, for instance for every P known to be non-universal.

Lemma 6.1. *For $p \in \mathbb{R}^d$ and $E \subseteq \mathbb{R}^d$, the pinned distance set $\Delta_p(E)$ contains no affine copy of $\mathcal{G}_q$ if and only if for all $x \in \mathbb{R}$ and $s \neq 0$ there is $n \geq 1$ such that the sphere of radius $x + sq^n$ about p is disjoint from E.* *[Lean: `not_containsGeomCopy_pinnedDistSet_iff`]*

Proof. $r \in \Delta_p(E)$ if and only if the sphere of radius r about p meets E. □

7. Formalization

The Lean development accompanying this note is available at https://github.com/haraldschilly/math-distance-set-similarity. It is pinned to Lean 4 `v4.34.1` and Mathlib commit `d13f23b`, the same versions as OpenAI's library, so the input statements typecheck identically. The four hypotheses of Section 1.4 are Lean propositions in `Statements.lean`. `FalconerStatement` and `DyadicAvoidanceStatement` are copied verbatim from OpenAI's statement files `FalconerAllDimensions.lean` and `DyadicAvoidance.lean`. Table 1 lists every result of this note with its Lean name and its hypotheses. All names are in the namespace `DistanceSimilarity`, and every listed theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. For example, Theorem B reads:

```
theorem pinned_dyadic_obstruction (hDy : DyadicAvoidanceStatement) {d : ℕ}
    (hd : 1 ≤ d) {η : ℝ} (hη0 : 0 < η) (hη1 : η < 1) :
    ∃ K : Set (EuclideanSpace ℝ (Fin d)),
      IsCompact K ∧ K ⊆ closedBall 0 1 ∧
      (0 : EuclideanSpace ℝ (Fin d)) ∈ K ∧
      ENNReal.ofReal ((1 - η) ^ d) * volume (ball (0 : EuclideanSpace ℝ (Fin d)) 1)
        ≤ volume K ∧
      0 < volume K ∧ dimH K = d ∧
      ¬ ContainsDyadicCopy (pinnedDistSet 0 K) ∧ ContainsDyadicCopy (distSet K) ∧
      (2 ≤ d → ∀ p, p ≠ 0 →
        (∃ a b : ℝ, a < b ∧ Icc a b ⊆ pinnedDistSet p K) ∧
        ContainsDyadicCopy (pinnedDistSet p K))
```

Re-checking the inputs. A script in the repository copies, from a checkout of `github.com/openai/math` at commit `adc7f12`, the transitive closure of OpenAI's Lean modules proving the two inputs: 2459 files, depending only on Mathlib. It also copies the Lean proof of the Mattila–Sjölin theorem [19] at commit `435cd62`; the OpenAI modules that proof uses are already among the 2459. All of this is built with the same toolchain. A further module states `theorem falconer_holds : FalconerStatement := OAI.Falconer.falconer_distance_conjecture`, and analogously `dyadicAvoidance_holds`

Result	Lean name	Hypotheses
Theorem 2.1	`falconer_holds` (in `verify/`)	proved by OpenAI
Theorem 2.2	`dyadicAvoidance_holds` (in `verify/`)	proved by OpenAI
Theorem 2.4	`mattilaSjolin_holds` (in `verify/`)	proved in [19]
Lemma 3.1	`eventually_volume_pattern_pos`	none
Theorem A	`finite_pattern_distSet`, `exists_finite_pattern_distSet`	Falconer
Proposition 3.2	`volume_pattern_pairs_pos`	Falconer
Lemma 3.3	`volume_distSet_inter_Icc_pos`	Falconer
Proposition 3.4	`exists_dyadicFree_pos_every_scale`	dyadic avoidance
Lemma 4.1	`not_containsGeomCopy_insert_zero`	none
Lemma 4.2	`volume_norm_preimage_ge`	none
Lemma 4.3	`dimH_eq_of_volume_pos`	none
Lemma 4.4	`exists_Ico_subset_distSet`	none
Lemma 4.5	`Icc_subset_pinnedDistSet`	none
Theorem B	`pinned_dyadic_obstruction`	dyadic avoidance
Theorem 4.6	`pinned_geometric_obstruction`	GA(q)
Proposition C(i)	`distSet_containsAffineCopy_of_interior`	none
Proposition C(ii)	`distSet_containsAffineCopy_of_volume_pos`	none
Proposition C(iii)	`counterexample_properties`	Falconer
Remark 5.1	`window_of_mattilaSjolin`	Mattila–Sjölin
Lemma 6.1	`not_containsGeomCopy_pinnedDistSet_iff`	none

TABLE 1. Results of this note and their Lean formalizations.

and `mattilaSjolin_holds`. It then derives unconditional versions of Theorem A, Proposition 3.2, Lemma 3.3, Proposition 3.4, Theorem B, Proposition C(iii) and Remark 5.1. The kernel accepts these, and `#print axioms` reports only the three standard axioms. We additionally replayed the entire resulting environment (Mathlib, OpenAI's modules, the Mattila–Sjölin proof and ours) in a fresh kernel with Lean's `leanchecker`. We did not run OpenAI's `comparator` setup. A formal proof certifies the statement as formalized. Readers should still compare the formal statements with the informal ones; to make this easy, the repository contains a plain-language guide to reading the Lean statements.

Acknowledgments and use of AI

This note builds on results produced by an AI model and released by OpenAI [15, 14, 16]. It was itself prepared with substantial assistance from Claude (Anthropic), an AI system, which helped choose the combination studied here, search the literature, draft the text, and write the Lean formalization, including the Lean proof of the Mattila–Sjölin theorem [19]. The author checked the mathematics and the formal statements and takes full responsibility for the content.

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